

19/08/2025

CUBICLE 5:

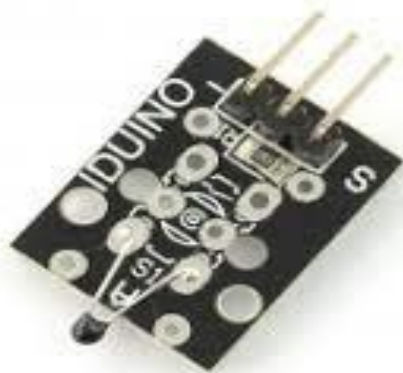
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Hall Effect Sensor Switch Module



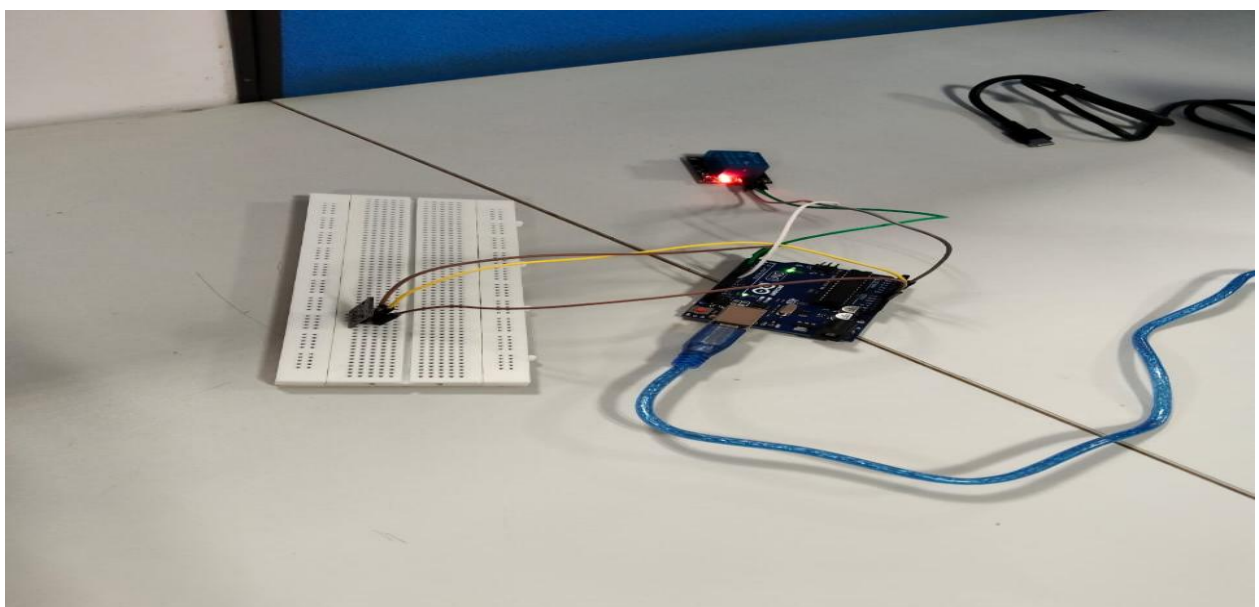
DESCRIPTION:

The Hall Effect sensor is an electronic switch that detects the presence of a magnetic field. When a magnet comes close, the sensor changes its output (ON/OFF). It works without touching, so it is often used as a contactless switch.

CONNECTION:

Module / Part	Pin Name	Connects To (Arduino Pin)
HW-492 Sensor	VCC	5V
	GND	GND
	OUT	D2
Relay Module	VCC	5V
	GND	GND
	IN	D8
Arduino UNO Pins	D2	Sensor OUT
	D8	Relay IN
	5V	Sensor VCC + Relay VCC
	GND	Sensor GND + Relay GND

BOARD SETUP:



PROGRAM:

```
int hallSensor = 2; // HW-492 OUT pin connected to D2
int relayPin = 8; // Relay IN pin connected to D8

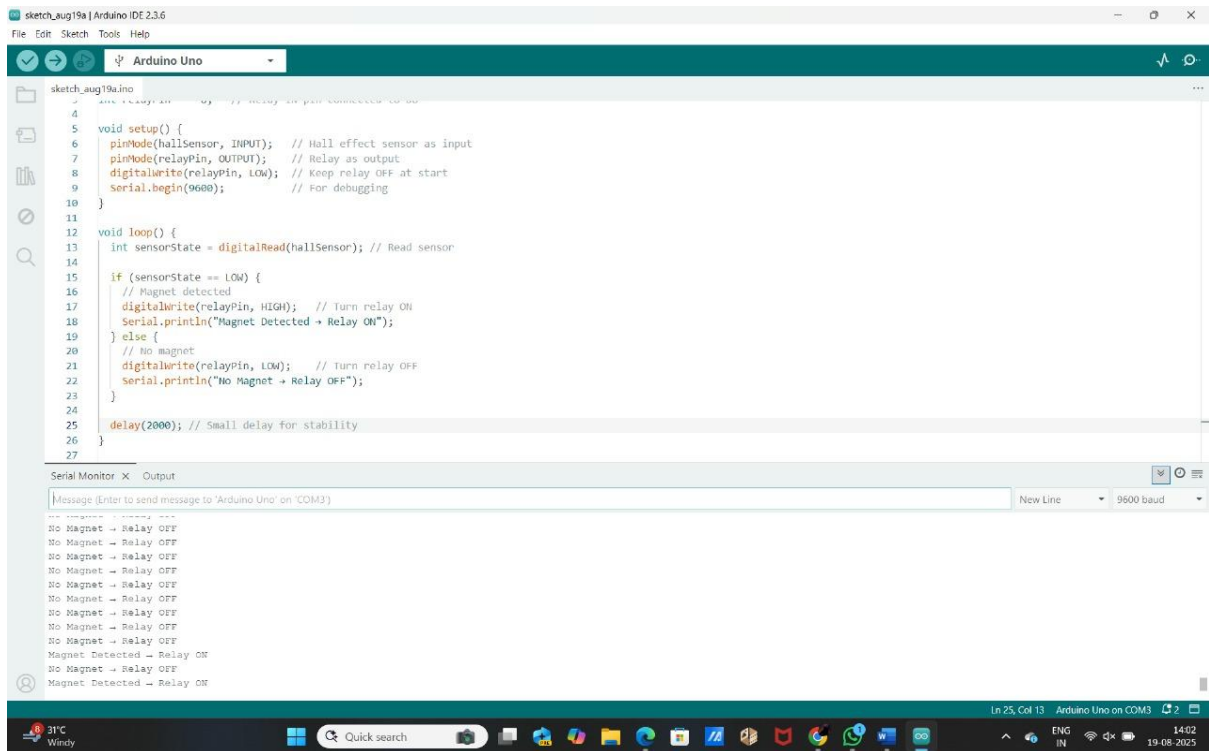
void setup() {
  pinMode(hallSensor, INPUT); // Hall effect sensor as input
  pinMode(relayPin, OUTPUT); // Relay as output
  digitalWrite(relayPin, LOW); // Keep relay OFF at start
  Serial.begin(9600); // For debugging
}

void loop() {
  int sensorState = digitalRead(hallSensor); // Read sensor

  if (sensorState == LOW) {
    // Magnet detected
    digitalWrite(relayPin, HIGH); // Turn relay ON
    Serial.println("Magnet Detected → Relay ON");
  } else {
    // No magnet
    digitalWrite(relayPin, LOW); // Turn relay OFF
    Serial.println("No Magnet → Relay OFF");
  }

  delay(100); // Small delay for stability
}
```

OUTPUT:



The screenshot shows the Arduino IDE interface with a sketch named 'sketch_aug19a' for an Arduino Uno. The code is as follows:

```
1 // Relay pin connected to pin 5
2 // Hall effect sensor connected to pin 4
3
4
5 void setup() {
6   pinMode(hallSensor, INPUT); // Hall effect sensor as input
7   pinMode(relayPin, OUTPUT); // Relay as output
8   digitalWrite(relayPin, LOW); // Keep relay OFF at start
9   Serial.begin(9600); // For debugging
10 }
11
12 void loop() {
13   int sensorState = digitalRead(hallSensor); // Read sensor
14
15   if (sensorState == LOW) {
16     // Magnet detected
17     digitalWrite(relayPin, HIGH); // Turn relay ON
18     Serial.println("Magnet Detected -> Relay ON");
19   } else {
20     // No magnet
21     digitalWrite(relayPin, LOW); // Turn relay OFF
22     Serial.println("No Magnet -> Relay OFF");
23   }
24
25   delay(2000); // Small delay for stability
26 }
27
```

The Serial Monitor shows the following output:

```
Message (Enter to send message to 'Arduino Uno' on 'COM3')
New Line 9600 baud
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
No Magnet -> Relay OFF
Magnet Detected -> Relay ON
No Magnet -> Relay OFF
Magnet Detected -> Relay ON
```

The bottom status bar indicates 'Ln 25, Col 13' and 'Arduino Uno on COM3'. The system tray shows a temperature of 31°C, a search bar, and the date/time 14:02 19-08-2025.